✅ Step 1: Enable GPU (once)

Runtime → Change runtime type → GPU

Verify:

```
!nvidia-smi
```

✅ Step 2: Import Libraries

```
from numba import cuda
import numpy as np
```

✅ Step 3: “Hello Kernel” CUDA Program

(2 blocks × 4 threads = 8 threads total)

```
@cuda.jit
def hello_kernel():
    # Thread and block indices
    tx = cuda.threadIdx.x
    bx = cuda.blockIdx.x
    bdim = cuda.blockDim.x

    # Global thread ID
    gid = bx * bdim + tx

    print("Hello from Block", bx, "Thread", tx, "Global ID", gid)
```

✅ Step 4: Launch Kernel

2 blocks, 4 threads per block

```
blocks = 2
threads_per_block = 4

hello_kernel[blocks, threads_per_block]()
cuda.synchronize()
```

✅ Expected Output (Order May Vary)

```
Hello from Block 0 Thread 0 Global ID 0
Hello from Block 0 Thread 1 Global ID 1
Hello from Block 0 Thread 2 Global ID 2
Hello from Block 0 Thread 3 Global ID 3
```

```
Hello from Block 1 Thread 0 Global ID 4
Hello from Block 1 Thread 1 Global ID 5
Hello from Block 1 Thread 2 Global ID 6
Hello from Block 1 Thread 3 Global ID 7
```

📌 **Important**: Output order may change because GPU threads run in parallel.

CUDA Execution Model

```
Grid
 ├── Block 0 → 4 threads
 └── Block 1 → 4 threads
```

Key CUDA Concepts

Concept	Value
Blocks	2
Threads per block	4
Total threads	8
Global thread ID	`blockIdx.x * blockDim.x + threadIdx.x`